

Data Architecture

Data Architecture Summary

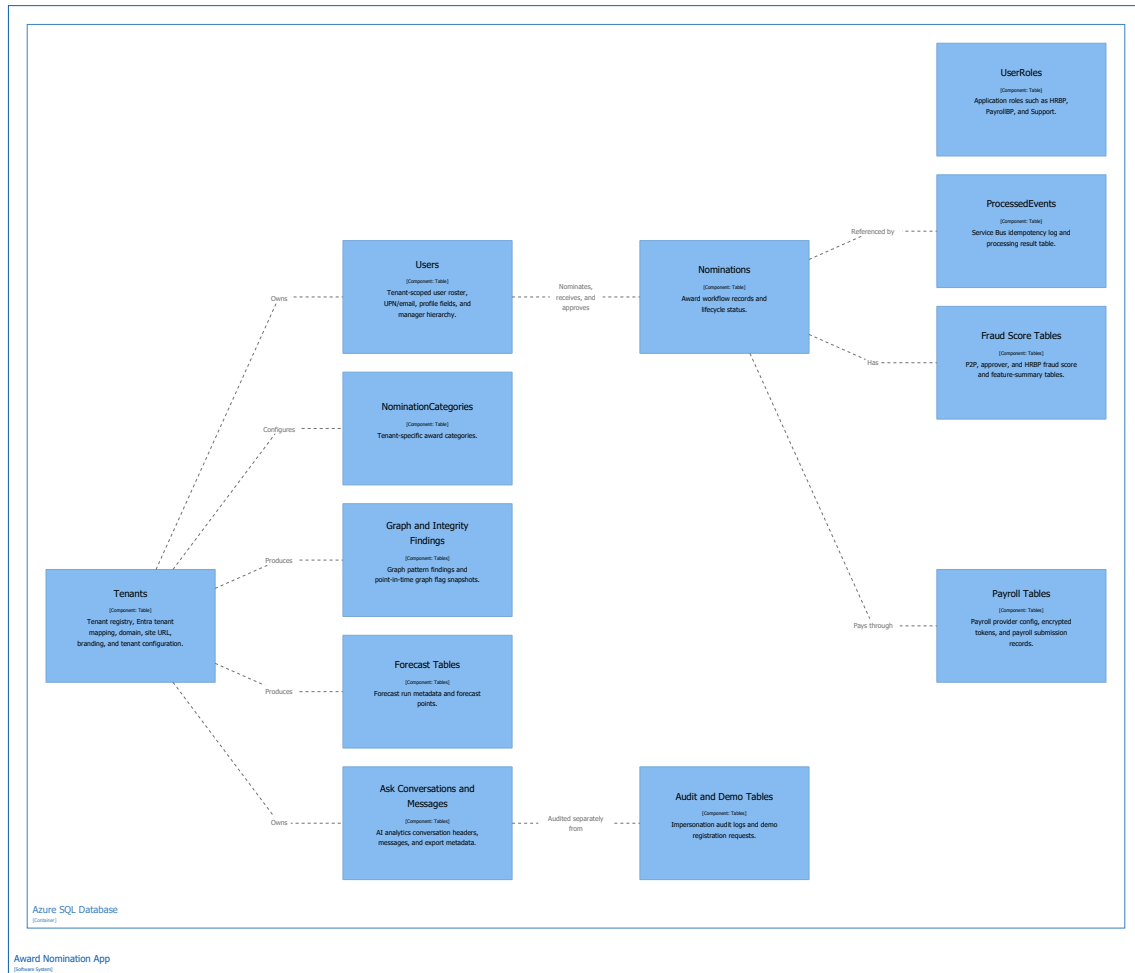
Azure SQL is the primary system of record. Blob Storage stores generated and binary artifacts such as fraud models, certificates, certificate templates, and exports. Service Bus carries event envelopes but does not own long-term business state.

The data model supports:

- Multi-tenant identity and configuration.
- User roster and manager hierarchy.
- Nomination lifecycle.
- Fraud scoring and HRBP review.
- Graph analytics.
- Forecasting.
- Notifications and idempotent event processing.
- Conversations for AI analytics.
- Payroll provider integration.
- Demo registration.

Conceptual Data Model

Structured source: [../diagrams/structurizr/workspace.dsl](#), view [ConceptualDataModel](#).



Component View: Award Nomination App - Azure SQL Database
Conceptual data model represented as Azure SQL table groups.
Tuesday, July 7, 2020 at 3:48 PM Pacific Daylight Time

Conceptual data model

Core Tables

Table	Purpose
Tenants	Tenant registry, Entra tenant mapping, domain, site URL, branding, config, certificate config, description config, integrity config, demo flag.
Users	Tenant-scoped user roster, UPN/email, profile fields, manager hierarchy.
UserRoles	Application roles such as HRBP, PayrollBP, Support with tenant support.
Nominations	Award nomination lifecycle record.
NominationCategories	Tenant-specific award categories.
ImpersonationAuditLog	Admin impersonation audit events.
ProcessedEvents	Service Bus idempotency and result tracking.
P2P_FraudScores	Nominator-to-beneficiary fraud scores.
Appr_FraudScores	Approver-pattern fraud scores.
HRBP_FraudFlags	HRBP review detail and feature summaries.
GraphPatternFindings	Weekly graph and semantic integrity findings.

Table	Purpose
UserGraphFlags	Point-in-time user graph feature snapshots.
ApproverPairFlags	Point-in-time approver pair feature snapshots.
ForecastRuns	Forecast run metadata and metrics.
Forecasts	Forecast points by series, level, grain, and horizon.
EmailTemplates	Tenant/language-specific notification templates.
Holidays	Holiday calendar data used by forecasting and analytics.
AskConversations	AI analytics conversation headers.
AskMessages	AI analytics message history and export metadata.
DemoRegistrationRequests	Demo access request audit and rate-limiting data.
payroll_providers	Tenant payroll provider configuration.
payroll_tokens	Encrypted provider OAuth tokens.
payroll_submissions	Payroll submission status for approved nominations.

Nomination Data Model

Important fields:

- NominationId
- TenantId
- NominatorId
- BeneficiaryId
- ApproverId
- DollarAmount
- Currency
- NominationDescription
- NominationDate
- Status
- ApprovedDate
- PayedDate
- ApproverNotifiedAt
- PaymentRef
- PaymentSubmittedAt
- RejectionReason
- RejectionActor
- CategoryId
- payroll_provider_id

Known status values:

- Submitted
- Pending
- PendingHRBPreview
- Approved
- Paid
- Rejected

Tenant Configuration Data

Tenant behavior is partly stored as JSON in **Tenants.Config** and related config columns.

Representative tenant config:

```
{
  "locale": "en-US",
  "currency": "USD",
  "min_award": 50,
  "max_award": 5000,
  "theme": {
    "primaryColor": "#4f46e5",
    "primaryHoverColor": "#4338ca",
    "primaryLightColor": "#e0e7ff",
    "primaryTextOnDark": "ffffff"
  }
}
```

Other tenant config areas:

- Description checks: word/character count, boilerplate phrases, embedding model, semantic thresholds.
- Certificate config: enabled, template blob, attach-to-beneficiary flag.
- Integrity config: routing thresholds and graph detection options.

Tenant Isolation Strategy

Tenant isolation uses application-level row filtering and identity resolution:

- Token **tid** maps to **Tenants.AzureAdTenantId**.
- Authenticated user must exist in **Users** for the resolved **TenantId**.
- API handlers pass **TenantId** into SQL helper calls.
- Analytics queries are tenant-scoped.
- HRBP and payroll access is checked through tenant-scoped roles.
- Domain mismatch is detected and surfaced in frontend behavior.

Future hardening option: add database row-level security policies for defense in depth.

Analytics Data Model

Analytics is derived mostly from:

- **Nominations**
- **Users**
- **P2P_FraudScores**
- **Appr_FraudScores**
- **HRBP_FraudFlags**
- **GraphPatternFindings**
- **ForecastRuns**
- **Forecasts**
- **UserGraphFlags**
- **ApproverPairFlags**

The analytics API returns aggregated shapes rather than exposing raw SQL tables.

Graph Data Model

SQL graph tables include:

- **NomGraph_Person**
- **NomGraph_Nominated**
- **NomGraph_NominationEmbedding**

The weekly graph job materializes detected findings and snapshots:

- **GraphPatternFindings**
- **UserGraphFlags**
- **ApproverPairFlags**

Patterns include:

- Ring.
- Super nominator.
- Nomination desert.
- Approver affinity.
- Copy-paste.
- Transactional language.
- Hidden candidate.

Data Lifecycle

Data	Creation	Update	Retention recommendation
Tenant config	Tenant onboarding	Admin/ops configuration	Retain for tenant lifetime plus contract period.
User roster	Seed/import/onboarding	HRIS sync or admin scripts	Retain while user is active; archive after departure per policy.
Nominations	User submission	Workflow status changes	Retain for audit and finance period, commonly 7 years if payout-related.
Fraud scores	Integrity worker and weekly job	Retraining/upsert	Retain with nomination records for explainability.
Graph findings	Weekly job	New runs	Retain rolling windows plus audit archive.
Event processing	Workers	Result updates	Retain operationally, e.g. 90-365 days.
AI conversations	Admin analytics	User updates/deletes	Retain per tenant privacy policy.
Payroll tokens	OAuth onboarding	Refresh	Encrypt at rest; remove on tenant offboarding.
Payroll submissions	Approval payout	Webhook/worker updates	Retain with payout audit period.

PII and Sensitive Data

PII includes:

- Names.
- UPN and email.
- Manager relationships.
- Payroll employee profile data.
- Address and payrate data returned through payroll lookup.
- Nomination text, which may include personal details.

Sensitive operational data includes:

- Payroll OAuth tokens.
- Webhook secrets.
- SQL credentials.
- OpenAI keys.
- Storage keys.

Controls:

- Key Vault-backed secrets.
- Token encryption for payroll tokens.
- RBAC and app roles.
- Tenant-scoped access.
- Audit logs for impersonation.
- Avoid exposing raw secrets in logs.